



Final Report

Web Technologies – ICT 1209

**Bachelor of Information and Communication Technology
(BICT)**

Degree Programme

Department of Information and Communication Technology

Faculty of Technology

Rajarata University of Sri Lanka

Mihintale

Details of the Project

Project Title : Routinefy – A Habit Tracker

Group Number : 16

Github repository link : <https://github.com/indun-art/routineify-habit-tracker>

Submission Date : 26/09/2026

Details of the Group Members

Name	Registration ID	Index No.
M.M.I.U. BANDARA	ITT/2024/021	2706
K.R.I.A. BANDARA	ITT/2024/019	2704

1. Introduction

Routinefy is a web-based habit tracking and management system developed using PHP, MySQL, HTML, CSS and JavaScript. The main idea behind the project is simple: users should be able to create habits, set targets, mark daily progress and check their performance without having to maintain records manually.

The system also includes an administrator area for managing users, habits, system settings and user feedback. During development, attention was given to login security, session handling, role-based access and database validation so that the system behaves like a small real-world web application.

2. Project Overview

2.1 Purpose

The purpose of Routinefy is to provide a convenient digital method for building and monitoring daily habits. Instead of keeping progress in notebooks or separate applications, users can keep their habit information in one place and view their progress through the dashboard and analytics pages.

2.2 Target Audience

The main target users are students and young adults who want to maintain regular routines such as studying, exercising, reading, drinking enough water or following other personal goals. The system can also be used by anyone who needs a simple personal habit tracker. Administrators are a secondary user group because they are responsible for managing accounts and monitoring system activity.

2.3 Key Features

Feature	Description
User registration and login	Users can create accounts and securely log in to the system.
Habit management	Users can add habits, set a target in days, mark a habit as completed and delete habits.
Daily completion tracking	The system records whether a habit was completed on a particular date.
Streak calculation	The dashboard calculates the user's current consecutive completion streak.
Analytics	Users can view completion statistics, highest streak, weekly activity and monthly activity.
Admin dashboard	Administrators can view system statistics, users, habits and feedback.
User management	Administrators can update roles and account status and remove user accounts where appropriate.
Password recovery	The system provides an OTP-based password reset flow.
System settings	Administrators can update the application name and

	registration status.
Feedback	Logged-in users can submit feedback from the dashboard; feedback is saved for the administrator and can be sent by email when SMTP/mail is configured.

3. System Design

3.1 Database Schema

The database is named routinefy and uses MySQL. The design separates users, habits and daily habit records so that one user can own several habits and each habit can have many completion records. A separate messages table is used for contact messages and user feedback.

Table	Main Fields	Purpose
users	id, full_name, email, password, role, status, reset_otp, reset_expires, created_at	Stores user and administrator accounts.
habits	id, user_id, name, goal_days, created_at	Stores habits created by users. user_id links to users.
habit_logs	id, habit_id, date, completed	Stores daily completion status for each habit.
messages	id, name, email, message, created_at	Stores contact messages and dashboard feedback.
settings	setting_key, setting_value	Stores configurable system values such as application name and registration status.

The main relationships are: one user can have many habits, and one habit can have many habit_logs. The foreign keys use cascading deletion so that related habit records are removed when a user or habit is deleted.

3.2 Folder Structure

```
Routinefy_PHP_MySQL_Final/
├── index.php
├── dashboard.php
├── analytics.php
├── admin-dashboard.php
├── manage-users.php
├── manage-habits.php
├── settings.php
├── signup.php
├── forgot-password.php
├── reset-password.php
├── contact.php
├── logout.php
├── database.sql
├── api.php
├── auth/
│   ├── login.php
│   ├── logout.php
│   └── register.php
├── includes/
└── db.php
```

```
|   └─ functions.php
├─ js/
|   └─ api.js
├─ css/
|   └─ style.css
|   └─ style2.css
└─ images/
    └─ background.png
```

The project uses a simple structure so that database connection and reusable PHP functions are kept inside includes/, front-end JavaScript is separated into js/, and styling is maintained in CSS files.

4. Implementation

4.1 Authentication and Role Management

The login process checks the email and password against the database and creates a PHP session after successful authentication. Administrator login has an additional security-key check. Page protection prevents normal users from directly accessing administrator pages.

4.2 User Dashboard and Habit Management

After logging in, a user is taken to the dashboard. The dashboard displays the user's name and current habits. A new habit can be added with a target number of days. The user can then mark the habit as done for the current day or delete it. The system stores these actions in the database rather than relying only on browser-side data.

4.3 Streaks and Analytics

Every completion is stored with a date in habit_logs. The system uses those dates to calculate consecutive completion streaks. The Analytics page uses database records to calculate the total number of habits, highest streak, completion rate, weekly activity and monthly activity.

4.4 Administrator Dashboard

The administrator dashboard provides an overview of total users, active habits, daily completions and feedback/messages. It also provides links to user management, habit management and system settings.

4.5 Feedback System

A feedback form was added to the bottom of the user dashboard. The logged-in user's name and email are taken from the active session, so the user does not need to enter them again. The message is stored in the messages table and is displayed in the Admin Dashboard under User Feedback & Messages. The project also contains email delivery through PHP mail(); for a local XAMPP installation, SMTP/mail configuration is needed for actual email delivery.

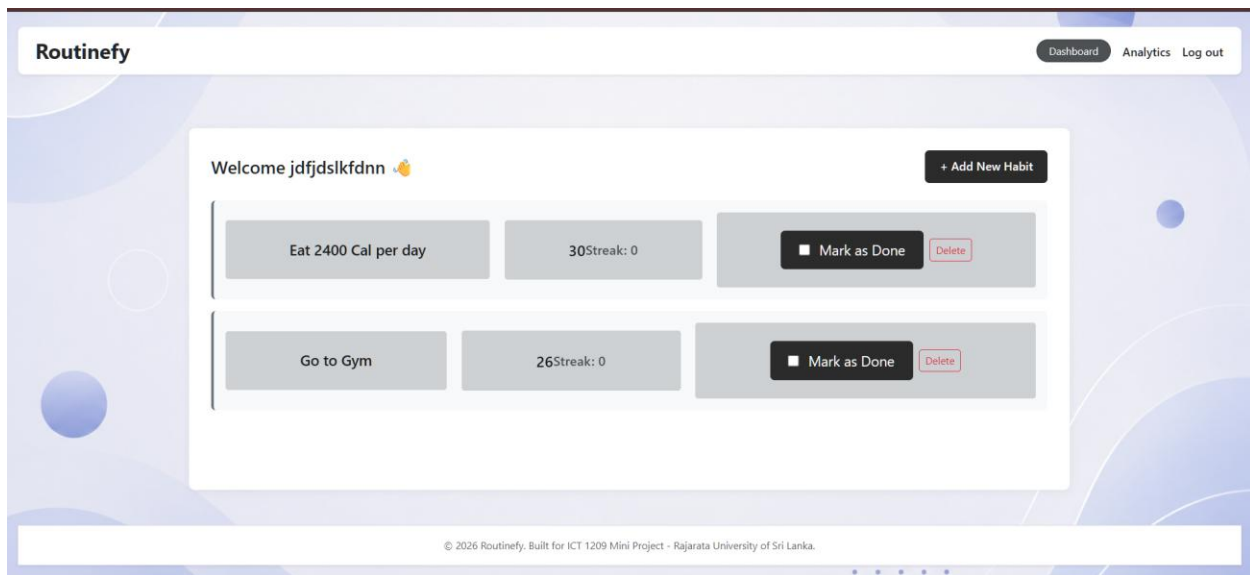


Figure 1. User dashboard showing the Routinefy habit tracking interface.

Additional screenshots recommended for the final submission are shown below. These should be captured after running the system so that they represent the actual project version used for the demonstration.

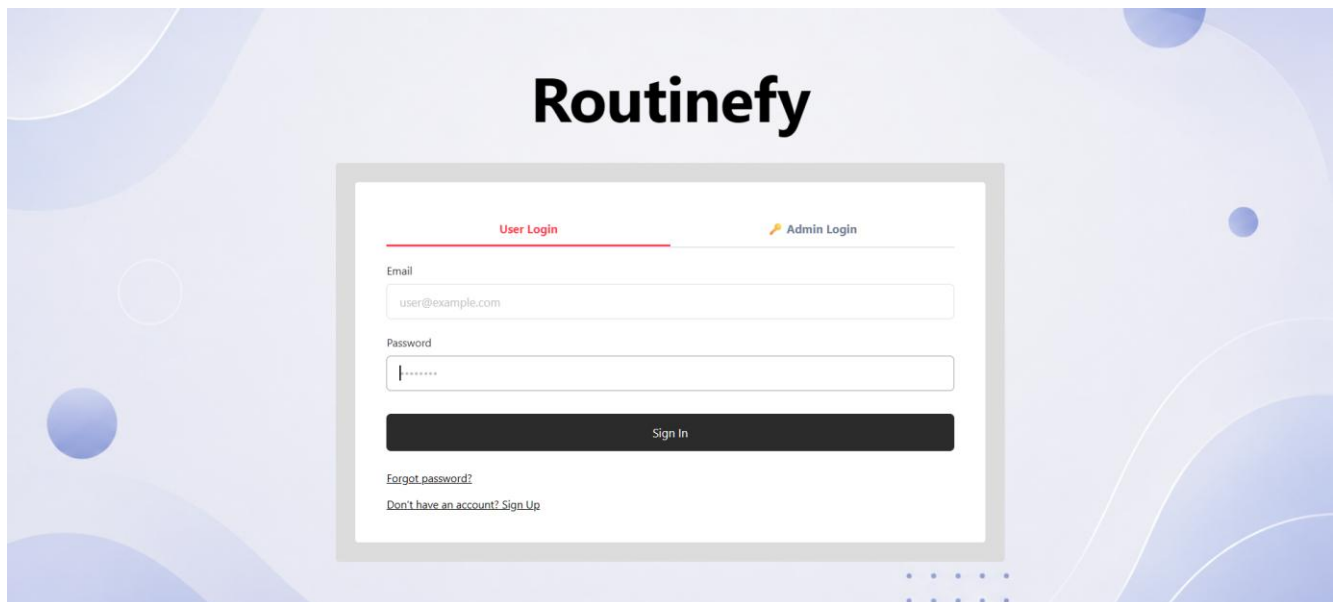


Figure 2 Sign In page

Routinefy

Full Name

Value

Email

Value

Create Password

Value

Create Account

[Already have an account? Sign In](#)

Figure 3 Create Account

Routinefy

Reset Your Password

Enter your registered email to generate a verification code.

Email Address

Send Verification Code

[Back to Sign In](#)

Figure 4 Forgot Password

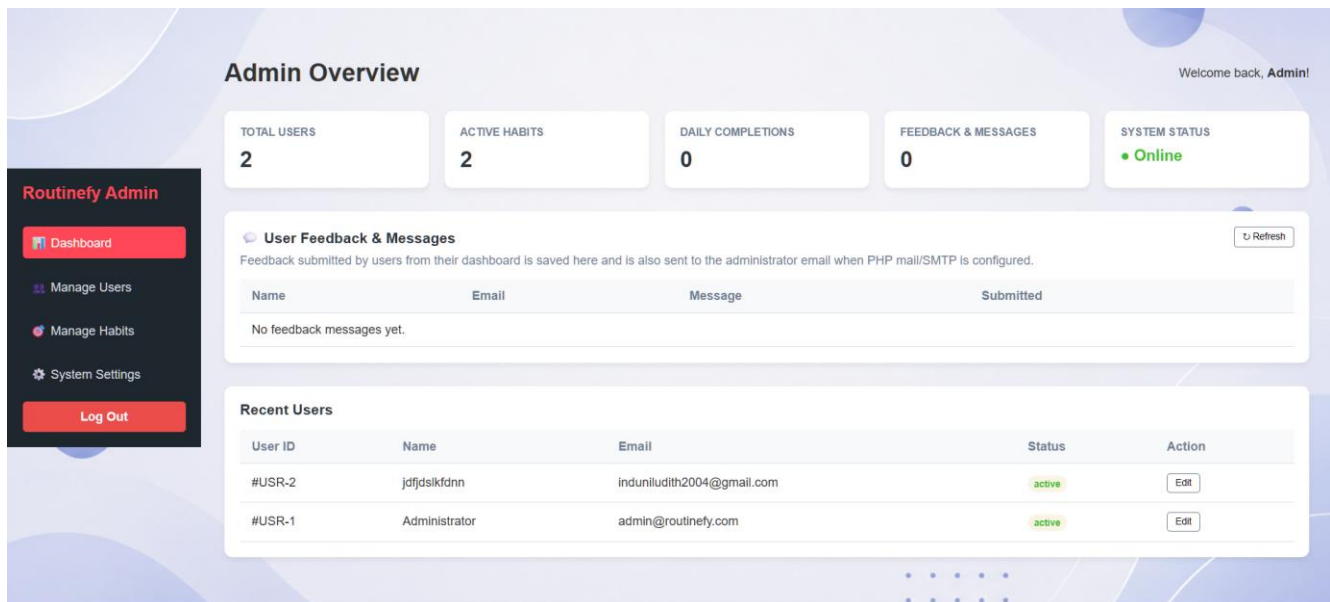


Figure 5 Admin Dashboard

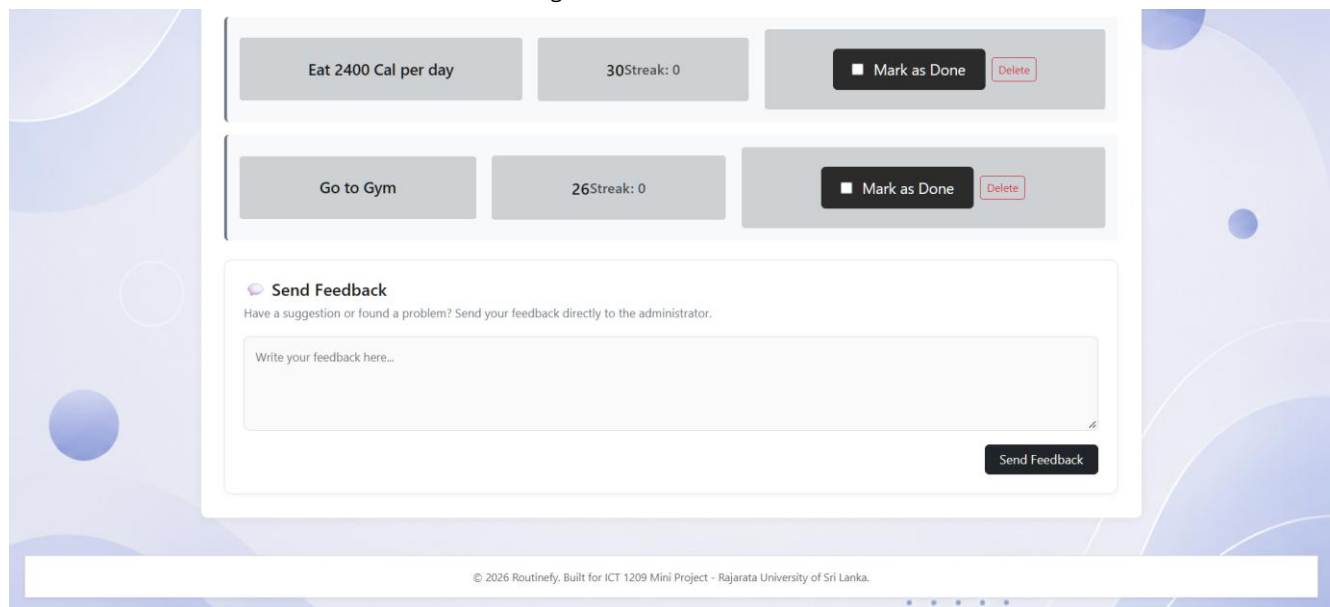


Figure 6 Feedback Section

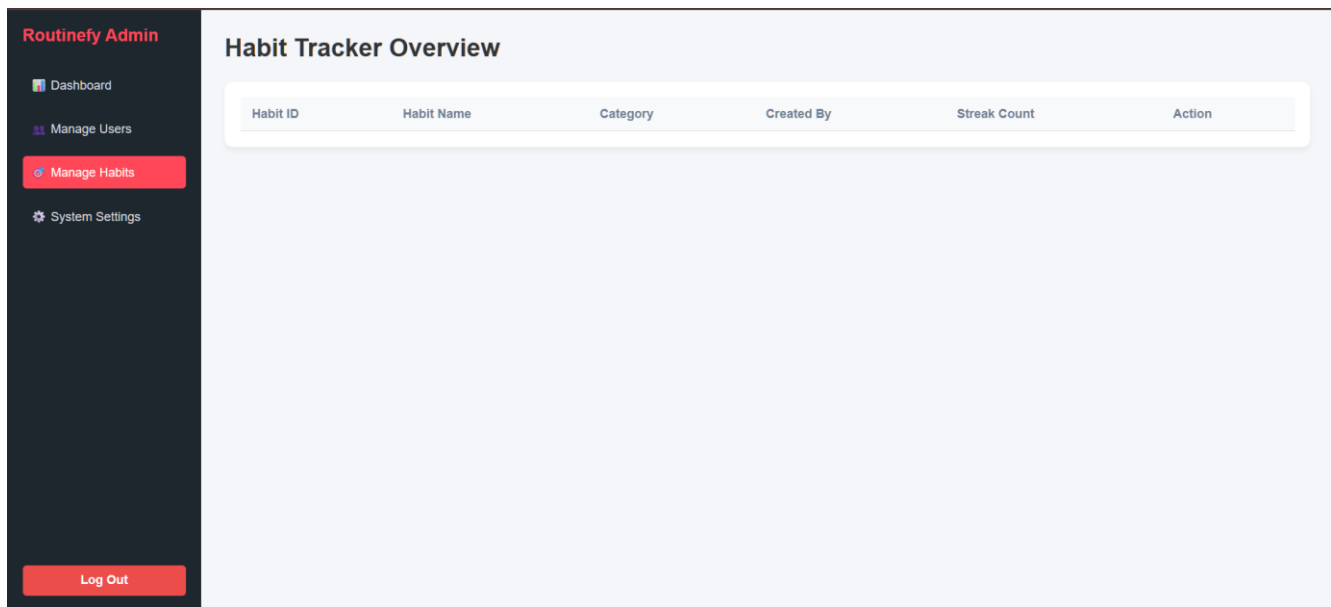


Figure 7 Habit Tracker Overview

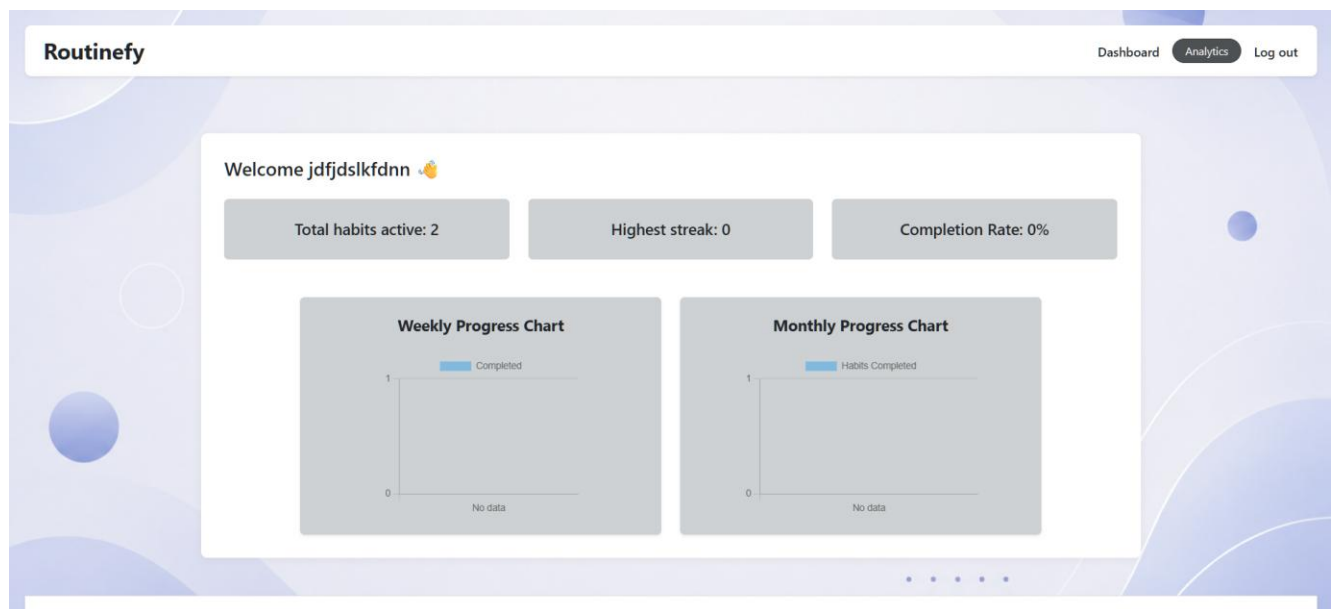


Figure 8 Analytics

5. Challenges and Solutions

Challenge	Problem	Solution
Role-based access	Users and administrators should not see the same pages.	Session checks and role checks were added to protected API routes and pages.
Keeping habit progress accurate	The same habit may be completed on different dates, so a single value	A separate habit_logs table stores one record per habit and date.

	is not enough.	
Correct streak calculation	A total completion count is not the same as a consecutive streak.	The application checks completion dates in reverse order to calculate the current consecutive streak.
Dynamic dashboard data	Hard-coded values would not reflect real user activity.	Dashboard and analytics values are loaded from MySQL through API endpoints.
Feedback handling	Feedback needs to be visible to the administrator instead of disappearing after submission.	Feedback is stored in the messages table and loaded into the admin dashboard.
Email delivery on XAMPP	PHP mail does not automatically deliver mail on a normal local XAMPP installation.	The system keeps a database copy of every feedback message and uses PHP mail() when an SMTP/mail service is configured.
Input and access security	Directly trusting browser input could allow invalid data or unauthorized actions.	Server-side validation, prepared SQL statements, sessions and output escaping were used.

6. Individual Contribution

The following breakdown is written in a practical project-report format based on the four-member team. If the actual work was divided differently, the names/activities in this table can be edited before submission.

Member	Main Area	Contribution
Indunil	UI/UX and front-end work	Dashboard layout, form design, styling, page consistency and user-interface improvements.
	Administration and testing	Admin dashboard, user/habit management, feedback handling, system settings and overall testing/debugging.
Induwara	Database and back-end work	MySQL database structure, PHP database connection, CRUD operations and API/database integration.
	Core functionality	Habit creation, daily completion, streak calculation, analytics and related testing.

Although responsibilities were divided, the project was completed as a group. Members also helped each other during integration, testing and fixing errors before the final version was prepared.

7. Setup Instructions

1. Install XAMPP and start Apache and MySQL from the XAMPP Control Panel.
2. Extract the Routinefy project folder into C:\xampp\htdocs\.
3. Open phpMyAdmin using <http://localhost/phpmyadmin/>.
4. Create/import the database by selecting database.sql. The SQL file creates the routinefy database, tables and demo administrator account.
5. Check includes/db.php and make sure the MySQL host, database name, username and password match the local XAMPP setup.

6. Open the project in a browser using http://localhost/Routinefy_PHP_MySQL_Final/.
7. Register a normal user or use the provided administrator account for the admin area.
8. Test the main flow: login → add habit → mark habit as done → view analytics → submit feedback → open the admin dashboard and check the feedback.

7.1 Demo Administrator Account

Email	admin@routinefy.com
Password	admin123
Admin security key	admin

For a real deployment, the demonstration password and security key should be changed. On local XAMPP, feedback is still stored in the database even if email delivery has not been configured.

8. References

PHP Documentation Group. *PHP Manual: password_verify* [Z/OL].

<https://www.php.net/manual/en/function.password-verify.php>

PHP Documentation Group. *PHP Manual: Password Hashing* [Z/OL].

<https://www.php.net/manual/en/security.database.storage.php>

Oracle Corporation. *MySQL 8.0 Reference Manual* [Z/OL]. <https://dev.mysql.com/doc/>

Bootstrap Team. *Bootstrap 5 Documentation* [Z/OL]. <https://getbootstrap.com/docs/5.3/>

Chart.js Contributors. *Chart.js Documentation: Getting Started* [Z/OL].

<https://www.chartjs.org/docs/latest/>

Font Awesome. *Font Awesome Documentation* [Z/OL]. <https://fontawesome.com/docs>

Mozilla Foundation. *MDN Web Docs: Web Technology for Developers* [Z/OL].

<https://developer.mozilla.org/en-US/docs/Web>

Apache Friends. *XAMPP Installers and Downloads* [Z/OL]. <https://www.apachefriends.org/>

Conclusion

Routinefy was developed as a practical web application for managing personal habits and monitoring progress. The final system combines authentication, habit tracking, streaks, analytics and administrator functions in one application. The addition of the feedback feature also provides a direct communication channel between users and the administrator. Through the project, the team gained practical experience in PHP, MySQL, HTML, CSS, JavaScript, database design, session management, API-based communication and testing. The system can be further improved in the future by adding reminders, mobile support, richer analytics and a fully configured SMTP service for reliable email notifications.